8 (a) State what is meant by a photon.

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(b) X-rays for use in medical diagnosis are produced by accelerating charged particles towards a metal target.

(i) State the name of the charged particles.

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(ii) Describe how X-rays are formed at the target.

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(iii) For any given accelerating voltage, there is a range of wavelengths of X-rays produced that has a minimum value.

Explain, with reference to photons, why there is a range of wavelengths and why this range has a minimum value.

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(iv) Calculate the minimum wavelength of X-rays produced when the accelerating voltage is 52.0 kV.

minimum wavelength = ........................................................ m [3]

**(c)** X-rays are used to create an image of the internal structure of an object that is made from two types of material of similar thickness.

Describe and explain how a property of the materials determines whether or not the image produced has good contrast.

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[Total: 13]